System Design & High-Level Architecture (HLD) Interview Handbook

Subject: System Design & Distributed Systems | Author: Aanya Sharma (IIT Delhi / ACM Fellow) | 100% Free Campus Access

1. DISTRIBUTED SYSTEMS FIRST PRINCIPLES

• **CAP Theorem:** In a network partition (P), a distributed system must trade off between Consistency (C) — every read receives the most recent write — and Availability (A) — every non-failing node returns a non-error response.
• **PACELC Theorem:** If Partition (P), trade off Availability (A) and Consistency (C); Else (E), trade off Latency (L) and Consistency (C). Explains DynamoDB and Cassandra design tradeoffs.

2. SCALABILITY & CACHING STRATEGIES

• **Cache-Aside (Lazy Loading):** Application checks cache first. If cache miss: queries DB, writes to cache, returns result. Pros: resilient to cache failures. Cons: high miss latency, potential stale data.
• **Write-Through vs Write-Back:** Write-Through writes to cache and DB simultaneously (higher write latency, zero data loss). Write-Back writes to cache immediately and flushes asynchronously to DB (ultra low latency, small risk of data loss on crash).
• **Consistent Hashing:** Maps servers and keys to a 360-degree hash ring. Adding/removing a server only shifts K/N keys rather than remapping all keys. Virtual nodes ensure uniform traffic distribution.

3. DATABASE SCALING & PARTITIONING

• **Horizontal Sharding:** Splitting tables across physical database instances by shard key (e.g. user_id). Avoids hot partitions by using hash-based sharding.
• **Read Replicas & Master-Slave Replication:** Single master handles all write transactions; multiple read replicas serve read-heavy workloads (e.g. 95% reads, 5% writes). Mitigate replication lag with sticky sessions.